

# Towards Clinical Practice Reflection Based on Repurposed Administrative Data: Qualitative Study of Physicians and Surgeons

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## Abstract

Clinicians are expected to make use of data to reflect on their practice; this is reflected in its accepted role in mandatory Continuing Professional Development (CPD). One potentially valuable form of relevant data that hospitals routinely collect is clinical indicators (CI), such as readmissions and complications. This paper describes our design and qualitative evaluation of SEECI, an interface to enable clinicians to use CI data to see, understand, and reflect on their practice. Our think-alouds with ten hospital staff indicate that SEECI enabled meaningful insights, despite the challenging nature of repurposed data, notably for valuable peer comparisons. Our key contributions: (1) SEECI, a new interface to support individuals' reflection on their clinical practice, based on data from a set of repurposed CIs; and (2) demonstration of the value of SEECI for individuals to gain insights and perspectives, including making use of peer comparison.

## CCS Concepts

• Human-centered computing → Empirical studies in HCI.

## Keywords

Clinical Indicators, Reflective Practice, Physicians, Surgeons, Continuing Professional Development (CPD), Practice Analytics, Think-Aloud Protocol

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## 1 Introduction

Medical practitioners are expected to regularly assess their performance and identify areas for development and improvement throughout their professional lives to provide appropriate care to their patients. This is in line with calls for clinicians' formal Continuing Professional Development (CPD) to involve reflection on data about their performance [40]. Hospitals currently collect considerable data that could be valuable for such reflection and CPD. One form of such data includes routinely collected clinical indicator (CI) data such as length of stay, readmissions, and complications. The primary purpose of CI data is for *institutional* tracking of quality and safety (Q&S) measures for mandatory reports to government departments and health insurers. Our work aims to repurpose this data to support *individual* clinicians' reflective practice to meet CPD accreditation requirements [7].

There are several challenges in repurposing routine clinical indicator data for practice reflection and professional development. First, the data was originally for patient administration and billing; it often requires specialised expertise to access and extract reliable forms of the data, which may be further compromised by coding errors or incomplete clinical notes [24]. Second, there are no standards for presenting these data in ways that are interpretable and clinically meaningful for individual reflection. One key difficulty is to attribute performance accurately to the appropriate clinician. In addition, peer comparison is critical for interpreting the data meaningfully. Clinicians value the ability to compare their performance data and patient outcomes with their peers [8, 41]. For example, surgeons need to know their complication and readmission rates (for a particular procedure with comparable patients) fall within expected ranges. Studies have found the use of head-to-head comparison data in audit and feedback tools has seen improved quality of care across different specialties [18]. However, doing this fairly and meaningfully involves the challenges in the nature of the data, often with small numbers and a large impact of different patient risk profiles [7]. Clinicians also have concerns that the data will be used punitively, rather than formatively for learning and development. Hospital staff must feel safe in reviewing team performance data and receiving constructive feedback without fear of negative consequences [4]. Despite these challenges, clinicians consider that clinical indicator data could be valuable for reflection and ultimately

drive improvements in clinician performance and patient health outcomes [32].

Our work aimed to repurpose routinely collected clinical indicators in administrative datasets to support clinician practice reflection. We designed a new reflective practice interface based on real-world clinical indicators, called **SEECI**, pronounced “Ceecee”, as it enables doctors to “see” their “clinical indicators”. Our research questions were:

- RQ1: What are clinicians’ perspectives on the repurposed administrative data in SEECI?
- RQ2: How does SEECI support peer comparison?
- RQ3: What insights can clinicians gain from reviewing their clinical indicator data using SEECI?

We designed a two-phase think-aloud study to understand how physicians and surgeons would use SEECI to review data about their practice. First, we conducted think-alouds with senior hospital administrators and staff clinicians. We provided participants with the SEECI interface to review their real-world performance data. Participants were asked to use the dashboard to review data about their practice whilst “thinking aloud”. We then refined the user interface based on participant feedback. Then, we conducted follow-up think-aloud sessions and asked participants to complete tasks on three different dashboard screens. Our work makes two key contributions to the CHI community: (1) SEECI, a new interface to support individuals’ reflection on their clinical practice based on data from a set of repurposed CIs; and (2) demonstration of the value of SEECI for individuals to gain insights and perspectives, including making use of peer comparison.

## 2 Related Work

Reflection is considered a central component of medical education and practice, although there is no agreement on a single definition or model. Nguyen et al. [28] proposed a five-component model of reflection based on a conceptual analysis of the major definitions. One definition that is particularly relevant to clinicians, is the concept of the “reflective practitioner”, introduced by Donald Schön to describe a professional who engages in reflection as a response to novel situations [36]. Schön describes *reflection-in-action*: real-time readjustments and recalibrating to situations during the experience, and *reflection-on-action*: post-event analysis of situations and responses to change future actions. Schön’s model for reflective practice has been studied and built upon in healthcare [26, 27] and education [2, 35]. The principles of reflection in- and on- action have also appeared in the HCI literature [3, 34]. Baumer [3]’s 2015 review showed that a majority of HCI papers that define reflection use Schön’s model. More recently, DunnGalvin et al. [12] adapted the model to describe reflective practice in anesthesiology.

Reflective practice and lifelong learning are central to the Continuing Professional Development (CPD) programs that ensure practising clinicians continually self-assess, and enhance their knowledge and skills throughout their career [9]. In Australia and New Zealand, medical practitioners must complete professional development activities that involve “reviewing performance” and “measuring outcomes”, every year, to maintain their accreditation to practice. This shift from traditional education activities, e.g. conferences and courses, to data-driven practice reflection has also

occurred in Canada [19] and the United Kingdom [20]. Moreover, traditional CPD requirements can be seen by clinicians as impractical, irrelevant to their scope of practice, and often treated as an administrative “check-box” activity [10]. Our work aims to provide clinicians with a new form of meaningful data for reflection, aligned with CPD standards.

## 2.1 Routinely collected clinical indicator data

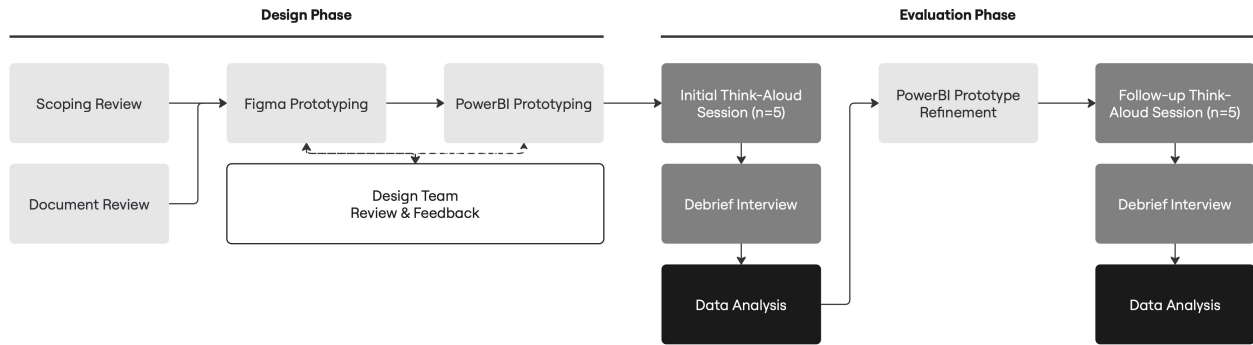
Hospital staff collect and generate data in Electronic Medical Records (EMRs) to deliver care to their patients. Clinical notes from these EMRs are then coded into the Patient Administration Systems used for patient management and billing purposes. Routinely collected clinical indicators can be extracted, such as length of stay, 28-day readmissions, and hospital-acquired complications, and are primarily used by hospital administrators and clinical informaticians to maintain top-down quality and safety standards [22]. Currently, individual clinicians and speciality care teams have limited access to these datasets for formative reflection on practice outside of traditional quality improvement tools, such as clinical audit [11] and mortality & morbidity (M&M) meetings [15]. There has been some recognition that performance data, such as these routinely collected clinical indicators, could support formative reflection and professional learning and development [39, 40]. We base our work on clinical indicator data because: it is widely available since hospitals collect these data for billing and reporting; clinicians and teams are familiar with the clinical indicators; and the indicators are used across multiple specialties.

## 3 Design of the SEECI interface

We designed SEECI to prompt participants to review their own data. Two main principles drove our design: (1) to select CIs that teams would be familiar with and value, and (2) to present the data using visual techniques familiar to clinicians and to support peer review. The design team included three researchers, with expertise in HCI, digital health and implementation science (authors 1, 5 and 6); two staff responsible for CI data reporting in a hospital (authors 2 and 3); and a clinician (author 4).

### 3.1 Design process

Our design process is shown in Figure 1. To select the clinical indicators for our interface, we drew from a scoping review [6]. We conducted a document review of the CIs in the *Hospital Casemix Protocol (HCP)* - a minimum dataset that hospitals are required to report to insurance companies and government departments [38]; and reviewed anonymised example *Excel* performance reports used in Orthopaedic Surgery at a research partner site based in Melbourne, Australia. We selected the CIs: length of stay, 28-day readmissions, hospital-acquired complications (HACs), operating time, unplanned returns to operating room, unplanned admissions to ICU, medical emergency team (MET) calls, and patient deaths. We also discussed the selection with team members (authors 2 and 3) who are responsible for CI reporting in our partner hospital. Our final selection was based on several factors, notably: importance; understandability; and generality of impact across medical speciality groups. The first author designed static wireframes using *Figma*. The first author shared these with the hospital members of the



**Figure 1: Design Process and Study Design.** The left side shows the design phase including static / interactive prototyping and design team review and feedback. The right side shows the evaluation phase including the two rounds of think-alouds and debrief interviews.

design team to refine them until there was a consensus to proceed with prototyping the designs. The first author then implemented the user interface of the dashboards in *PowerBI*, the platform the hospital deployed for hospital executive quality and safety monitoring. Finally, the first author met with each design team member separately to gather feedback to refine the design.

### 3.2 Description of the design

Figure 2 shows the *Your Trends* screen of the *SEECI* dashboard. We designed this to show the clinician’s data trends for: admissions (count); length of stay (days); hospital-acquired complications (count); and 28-day readmissions (count). These are presented together so that the clinician can take account of their caseload (admissions) when interpreting the three performance-related measures. We also provided data for their care team, even though we appreciated the challenges in the nature of this data. The interface enables click-through to case IDs so a clinician can dig into the details for a particular patient in the electronic medical records (EMR).

We designed the screen to use filters to give clinicians control over the data displayed so they could select data that is important to their practice. The *Speciality group* dropdown selects the speciality of the patient case - this is helpful when a clinician is affiliated with multiple speciality teams. The *Time period* dropdown filters for a particular year, quarter, or month. The *Procedure (MBS)* filters for a particular procedure or Medicare Benefit Schedule item number. The *Diagnosis Related Group (DRG)* dropdown filters for a group of patient episodes with similar diagnoses requiring similar hospital services. The *Doctor role* dropdown filters on their relationship to patient episodes, e.g. Original Admitting, Consulting, Principal Surgeon, or Assisting Surgeon. This will alter the number of patient episodes (admissions) displayed.

## 4 Methods

We conducted a two-phase think-aloud protocol (TAP) [13, 23]. The user study consisted of an initial think-aloud session, refinements to the *SEECI* user interface design based on participant feedback, and then a follow-up think-aloud session. Each think-aloud concluded

with a short debrief interview. The study setting was a large metropolitan private hospital in Sydney, Australia. The hospital provides a range of acute surgical, medical and obstetric care, including complex cardiac procedures, robotics and minimally invasive surgery – and supportive diagnostic and allied health services. Ethics approval to conduct the research study was granted by Adventist Health-Care Limited (AHCL) Human Research Ethics Committee (HREC) [2020-035]. We recruited a purposeful sample of senior hospital administrators, staff clinicians, and registrars (residents) [31]. Medical practitioners from any speciality were recruited. Purposeful sampling was used as clinicians are known to be overworked and time-poor. We aimed to recruit five participants for each round of interviews, in line with the iterative refinement of the user interface [29, 30].

### 4.1 Participants

Table 1 summarises the participants we interviewed in our study. Two of the ten participants (P01 and P02) were senior hospital staff who were champions for the project and had familiarity with the datasets for quality and safety reporting requirements. All but P02 had clinical experience. There was a balance between physicians and surgical specialists; their years of experience ranged from 8 to 28 years.

### 4.2 Procedure

In the initial set of think-alouds (P01 - P05), at the start of each session, the first author introduced the purpose of the dashboard. Participants were encouraged to explore each dashboard screen they were interested in. Participants were instructed to verbalise their thoughts and encouraged to ask clarifying questions [13]. The third author observed the sessions and kept detailed notes. After each session, the first and third authors met to debrief and compare notes. As P01 and P02 were not practising clinicians, they had no personal data. They were presented with the Head of Department user view of the dashboard, where they could see the performance data of all clinicians in a particular speciality group. Sessions were conducted online or in-person (as in the last column of Table 1) and ran for 30 to 45 minutes.

**Table 1: Think-Aloud participants.** For each participant, we report on their role within the hospital, sex, speciality / sub-specialities, number of years of experience, and whether the interview was online or in-person.

| PID | Role                      | Sex | Speciality                | Years experience | Session   |
|-----|---------------------------|-----|---------------------------|------------------|-----------|
| P01 | Chief Medical Officer     | F   | Medical Administration    | 7                | In-person |
| P02 | Chief Information Officer | M   | Technology Administration | 22               | In-person |
| P03 | Gastroenterologist        | M   | Physician                 | 21               | Online    |
| P04 | Urologist                 | M   | Surgeon                   | 16               | In-person |
| P05 | Medical Oncologist        | M   | Physician                 | 28               | In-person |
| P06 | Urologist                 | M   | Surgeon                   | 18               | Online    |
| P07 | Neurologist               | M   | Physician                 | 22               | Online    |
| P08 | Urologist                 | M   | Surgeon                   | 17               | In-person |
| P09 | Neurology (registrar)     | M   | Physician                 | 8                | Online    |
| P10 | Neurologist               | F   | Physician                 | 10               | In-person |

In the follow-up think-alouds (P06 - P10), our team decided to limit the session to three screens so that sessions would be around 30 minutes long (to be manageable for clinicians to complete them without interruption). The second and third authors chose the screens based on their experience in hospital administration. The screens were for: Overall Outcomes, 28-day Readmissions, and the Your Trends (Figure 2 above). The dashboard screens used in the think-aloud sessions are included in Appendix A. Participants also made use of the drill-down from these to the Case Details screen, which provided detailed information on a single patient case.

For each screen, the participant was given a task, e.g. “*Suppose you want to find your length of stay and operating room time relative to your peers, for a particular procedure in 2023. Please use the dashboard screen to do that*”. The think-aloud protocol tasks and debrief questions are included in Appendix B. Participants were reminded to verbalise their thoughts as they did the task. Finally, participants were asked to reflect on their experience in a short debrief interview. Participants were asked how important the clinical indicators were for them, what they learned from reviewing their clinical indicator data, and what changes, if any, would they make to their practice. Sessions ran for 26 to 44 minutes.

Each session was transcribed, de-identified, and thematically analysed [5]. The first author read each transcript to become familiar with the data while noting initial ideas, then transferred the text into a spreadsheet for open coding. An initial set of topic codes was generated by identifying the relevant features of each quote. All unique codes associated with quotes were consolidated into preliminary themes, which the first author reviewed with the last author in the context of the entire dataset. They refined the themes and then identified illustrative quotes to highlight each theme.

## 5 Results

### Clinicians highlighted challenges in interpreting and applying repurposed clinical indicators (RQ1)

*Learnability.* Overall, participants were familiar with the set of clinical indicators. However, some struggled with some unfamiliar terms. **P09** found the Complications screen difficult to use and did not understand what “HAC Rate” was or how it was calculated: “*I’m not sure how to interpret this graph exactly... I guess this is the case*

*numbers, but I’m not sure what the rate is referring to*”. They also needed clarification on how “28-day readmissions” were calculated: “*This graph doesn’t make a whole lot of sense to me and just the 28-day readmissions, it’s not clear to me at what point the patients got readmitted and what month*”. Participant comments indicated that broadly, they appreciated being able to see their performance data in SEECI. **P09** said: “*I like this screen for showing in a graphical form the admissions by diagnosis... it’s quite useful information*”. However, one participant commented on the volume of data presented and suggested using colour and adding more instructions about the drill down.

*Nature of the data and meaningfulness of the indicators.* Participants commented on the nature of the data. They highlighted differences between the data as collected and their needs for reflection and improvement. For example, **P06**, a surgeon, said: “*As a clinician, I want to know how much time I’m actually spending operating. From an admin perspective, you need to know what is the theatre time allocated for that particular item number and DRG (Diagnosis Related Group)*”. **P10** noted the dashboard was seen as a tool for driving department-wide initiatives and planning, such as identifying trends for resource allocation or creating new clinics based on patient needs.

Participants emphasised the importance of linking clinical indicators to clinical context, such as diagnostic reasons for readmissions and specific procedural outcomes, to make SEECI more actionable. **P07** highlighted how some indicators were not relevant for their team, such as the admissions data not matching their team’s unique workflow. Similarly, **P08**, a surgeon, commented on indicators that are useful for surgeons but likely not for others, particularly physicians. **P09** stressed the need for data and metrics matching requirements of Neurology, for example, focusing on relevant complications like delirium rather than surgical-related metrics.

Participants discussed how combined or multidisciplinary surgeries disrupted the neat categorisation of procedures. For Urology, **P08** explained the data attribution obstacles in obtaining accurate personal insights, such as linking procedures to specific surgeons or capturing multiple MBS procedure item numbers in grouped data. For non-surgical specialities like Neurology, **P07**, as noted above



Figure 2: Your Trends screen of the SEECI dashboard for a Urologist. To maintain the confidentiality of the actual data, we created synthetic performance data, based on an amalgam of several randomly chosen typical clinicians. The top left chart shows 12 months of admission counts for the clinician (red) compared to their team (grey). The top right chart compares their average patient length of stay (LOS) (red), in days, with their team (grey). The bottom left chart shows the number of hospital-acquired complications e.g. falls, infection, delirium. The HAC Group dropdown filters on a particular complication. The bottom right chart shows the number of unplanned readmissions of patients within 28 days of discharge.

highlighted a similar problem of attribution. P10 also referred to Neurology's shared care model, where multiple clinicians manage a patient across different phases of care.

### Peer comparisons foster meaningful self-assessment and benchmarking of clinical performance (RQ2)

Participants commented positively about SEECI's peer comparison, fostering self-assessment and benchmarking. P07 said: "It gives me an indication how my admissions data might compare to the admissions data of the Neurology department as a whole at the same time." P08 also valued peer comparison, but also awareness of possibility limitation, saying: "It looks like I'm better than average, which makes me feel good... but I wonder whether it's not picking up everything". Participants highlighted that this type of review of their data would contribute to their annual CPD requirements, P06 said:

"It's important from an audit perspective, because it's firm data points that are audited in our practice that can be comparable for the purposes of College of Surgeons CPD. So that's one thing. The second thing is, the

length of stay is very useful for us to know in terms of comparing how everyone else is going. It helps me understand whether what I'm doing is in line with everyone else and whether or not there's something I need to look at, you know, in my colleagues to figure out how to improve on." [P06-Urologist]

Notably, there were no comments about the problems of comparisons against clinicians with different caseloads.

Participants also commented on SEECI helping them review their performance data over different time periods, to compare their current outcomes with their past selves. P10 highlighted this would be useful for planning: "I guess in terms of planning. For my own schedule, it would be helpful to know the trends across the year with my on-call weeks to understand busier months and if there is a trend within that month". They also explained how this data would support the audit process of their practice in the future, P10 said:

"So, I might want to look at hospital complications for my weeks on call. So I've done three weeks on call this year. And I wonder if, I could have the data from hospital complications from my first week on call, implement a change, whether that be... give more stringent

*note taking, better coding, daily checking of the bloods, counselling the interns on maybe CT, whatever, and then comparing the data for subsequent weeks.” [P10-Neurologist]*

### Clinicians’ reflections on their performance data revealed new insights into their practice (RQ3)

All participants were able to reflect on their performance data and describe what they learned about their practice. Several reflections have already been reported above, such as **P08** on doing better than average on length of stay and its meaning for them. **P07** also commented on being within the “normal or expected range”. **P07** reflected on possible changes in their practice: *“I learned how my admission data compares to my peers. It might tell, certainly when it comes to the length of stay, if there are any obvious indicators that maybe patients are being discharged too early or that I’m hanging on to them for too long”*. Participants provided insights on internal work processes and the implications for data accuracy and completeness. **P06** remarked: *“I made some inferences about my practice. I learned some things about my practice, but I learned more about the challenges associated with keeping accurate data in complicated situations [...] I did learn things about my practice in that I’m not out of step in prostatectomy, which is a very straightforward thing”*.

## 6 Discussion

### Proactive, regular review of data to reduce quality of care issues

Our design took account of some inherent challenges in the nature of this data. Our findings underscore the challenges in repurposing a dataset that was originally collected for patient administration and billing. Some of these barriers relate to data quality and completeness [24], punitive use of the data by administrators [4], and accurate individual attribution of performance data in shared practitioner models of care [16]. Our work goes beyond the previous argument for the potential value of such data if it could be made available to clinicians [7, 37].

Clinicians we interviewed found reviewing this data to be most valuable for peer comparison and benchmarking. They found the ability to compare their current workload (admissions) and performance (complications, readmissions) with their past self was also useful. This type of proactive, regular review of performance data could potentially reduce the number of adverse events and improve the quality of care, as seen in studies on the impact of Clinical Decision Support Systems (CDSS) [14] and Audit & Feedback (A&F) tools [33].

### Potential to play a useful role in meeting annual CPD standards

Participants felt that SEECI may play a useful role in their Continuing Professional Development reflections. They noted having the ability to review their data regularly, data that was traditionally inaccessible, was seen as a trigger for self-reflection. There is a push from medical educators to integrate clinical performance data, such as routinely collected CIs, into CPD programs [39, 40]. However, they note integration to be limited due to similar issues we

identified, such as trustworthiness, data infrastructure, alignment concerns, and readiness of clinicians. Opening routinely collected hospital administrative data and repurposing these clinical indicators to support ongoing practice reflection would meet the new CPD standards and may also address some of the practical and conceptual challenges of CPD [10, 32].

### Risks and considerations for designing reflective interfaces based on repurposed data

The use of hospital data traditionally collected for administrative purposes has been combined and linked with other datasets to generate new insights into real-world clinical practice and patient outcomes [17]. While administrative datasets serve important billing and operational functions, our findings show that clinicians also want more granular, context and speciality-specific information that can meaningfully drive changes in practice, such as clinical notes from the EMR. The key benefit of using general process and outcome indicators (e.g. length of stay, 28-day readmissions, hospital-acquired complications) is that they are routinely collected in every hospital and have broad relevance to different speciality groups [25, 38]. These CIs can be used as a method to find “cases of interest” to trigger reflective discussions for learning and development [41]. However, care is needed when selecting CIs to account for differences in clinical contexts - such as distinct surgical and physician-patient workflows. Moreover, variations in patient co-morbidities, age, and disease diagnosis can influence patient outcomes. In the absence of risk-adjustment to control for these factors, clinicians can be unfairly benchmarked for treating higher-risk patients [21].

## 7 Limitations and Future Work

A key limitation is in our purposeful recruitment of participants, although we sought a balance of physicians and surgeons. This limits generalisability to other specialties and jurisdictions. Our design was influenced by the *PowerBI* platform, a forced choice based on its use in our hospital context. Even so, our work provides a foundation for a new class of interface for data-informed clinician reflection and CPD. Valuable future directions include: incorporating additional data sources (e.g., health records, billing) for CPD [40]; exploring how to facilitate deep understanding and sensemaking for reflection on medical practice [41]; and integrating feedback on CI data in useable ways to guide self-assessment [1].

## 8 Conclusion

Our work was driven by a vision for creating a new interface, called SEECI, to repurpose a set of real-world clinical indicators, extracted from routinely collected hospital administrative data that clinicians do not traditionally have access. We aimed to support data-informed practice reflection that would contribute to annual Continuing Professional Development requirements for clinicians to maintain accreditation and be a valuable support of lifelong learning. We conducted think-alouds with a diverse group of senior hospital staff, physicians, and surgeons. We found that the clinicians did see value in reusing these clinical indicators to support regular, formative practice reflection that would contribute to their CPD. They highlighted design considerations around the quality

of the underlying data, selection of indicators that are clinically meaningful to each speciality group, need for case-level details to uncover actionable insights, and accurate practitioner attribution in shared patient care models. These aspects need to be addressed to enable the adoption of these repurposed administrative datasets to support reflective practice.

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## A Dashboard Screens



Figure 3: *Overall Outcomes Screen*. The figure shows synthetic performance data. The first table shows one row as the specialty group “Urology”, with columns: admissions, average length of stay, complications, readmissions, unplanned returns to operating theatre, unplanned admissions to intensive care unit, medical emergency team calls, and deaths. The second table shows the same clinical indicators as columns. Each row shows the clinical indicators for each DrKey in the “Urology” group.

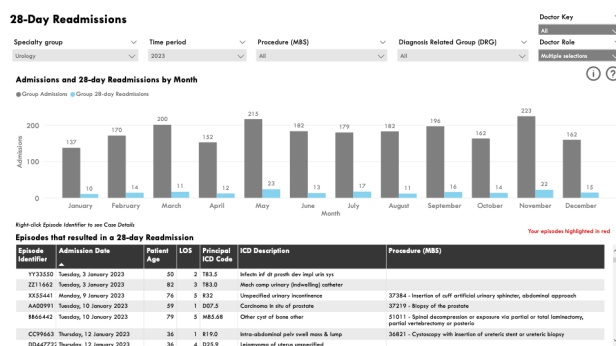


Figure 4: *28-Day Readmissions Screen*. The figure shows synthetic performance data. The top chart shows a clustered bar chart. The x-axis shows the months of the year. The y-axis shows the number of admissions. For each month, the bar chart shows two bars, the “Urology” group admissions (grey) and the group 28-Day Readmissions (light blue). The bottom table shows the patient episodes that have resulted in a 28-day readmission, with columns: episode identifier, admission date, patient age, length of stay, principal ICD code, principal diagnosis, and procedure.

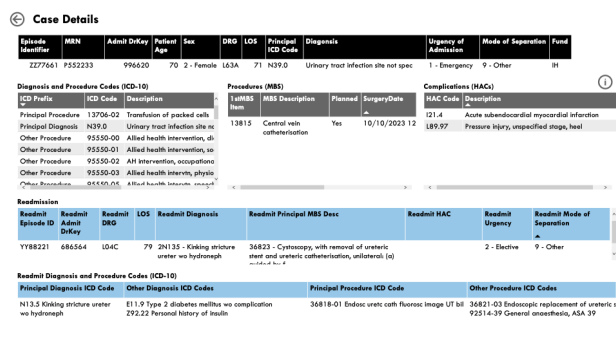


Figure 5: *Case Details Screen*. The figure shows synthetic performance data. The top table (black) summarises the patient’s episode. The next row of tables (grey) shows summary information for: diagnosis and procedure (ICD) codes, procedures (MBS), and complications (HACs). The last two tables (light blue) show summary information if the episode resulted in a 28-day readmission.

## B Think-Aloud Protocol

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### B.1 Tasks

- Task 1
  - **Material:** Outcomes Screen
  - **Description:** Suppose you want to compare Doctors in your group across multiple clinical indicators for the diagnosis “Stroke” in 2023. Then, find a particular Doctor to view all their Stroke cases. Please use the dashboard screen to do that.
- Task 2
  - **Material:** Your Trends Screen
  - **Description:** Suppose you want to compare trends in Admissions, Length of Stay, Complications and Readmissions with your fellow peers for the diagnosis “Stroke” in 2023. Please use this dashboard screen to do that.
- Task 3
  - **Material:** 28-Day Readmissions Screen + Case Details Screen
  - **Description:** Suppose you want to find patient episodes that resulted in a 28-day Readmission in 2023 for a particular diagnosis. Then, find one patient case you want to review in more detail. Please use this dashboard screen to do that.

### B.2 Debrief questions

- How valuable is [Clinical Indicator] to learn more about your practice? Please elaborate. (importance)
- By reviewing your [Clinical Indicator] data, what did you learn about your practice? (insight)
- What changes, if any, would you plan to implement into practice after reviewing your [Clinical Indicator] data? (goal setting, planning)
- What are the best things about this screen? Why? (usability)
- If you could change one thing about this screen, what would it be? Why? (usability)